

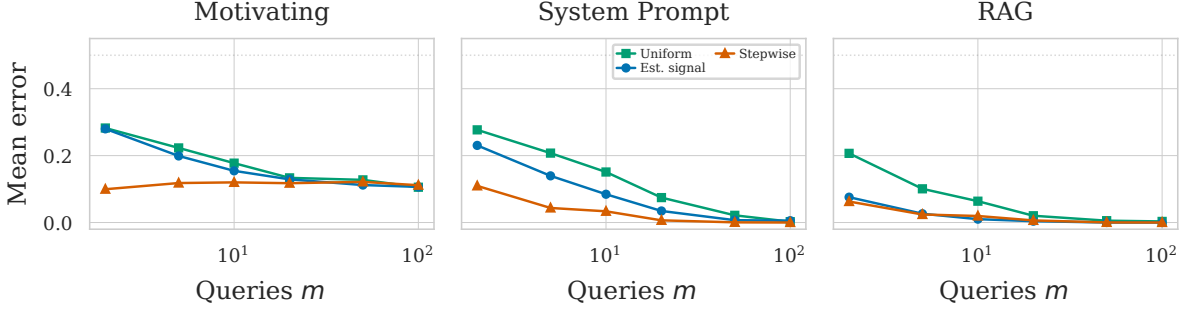


Figure 1. Mean classification error vs. query budget m for three selection strategies across three tasks ($n=80$). Stepwise selection matches uniform sampling accuracy with 5–10 $\times$ fewer queries.

1 Motivation

Classifying black-box generative models requires querying each model with m queries and comparing responses via pairwise energy distances. The total inference cost scales as $m \times n$, so reducing m while maintaining accuracy directly reduces per-model audit cost.

The discriminative factorization estimates per-query signal magnitudes $|\hat{\alpha}_{q,\ell}|$ from a labeled pilot set. We describe three strategies for using this information to select a targeted query set Q^* of size m .

2 Methods

1. Uniform Sampling (baseline). Sample m queries uniformly at random from the pool of M queries. No pilot information is used.

2. Estimated Signal Sampling. Fit a two-component GMM to the SVD loadings $|U_{q,\ell}|$ of the energy tensor E . Queries assigned to the higher-mean component form the estimated signal set $\hat{\mathcal{S}}$. Sample m queries uniformly from $\hat{\mathcal{S}}$. This exploits zero-set structure but ignores query redundancy.

3. Forward Stepwise Selection. Define a response $z_{ij} = \mathbf{1}[y_i \neq y_j]$ for each model pair and a predictor $E_{q,ij} = \|g(f_i(q)) - g(f_j(q))\|^2$ for each query q . At each step, add the query whose $E_{q,\cdot}$ has the largest partial correlation with the residual of z after projecting out previously selected queries. This is Gram-Schmidt orthogonalization on E w.r.t. z , ensuring each query contributes maximally non-redundant signal. Cost: $O(m \cdot M \cdot \binom{n}{2})$ — negligible.

3 Setup

Task	n	M	Description
Motivating	100	200	Sensitive training data
Sys. Prompt	100	200	Covert persuasion bias
RAG	120	200	Unauthorized doc. stores

$n_{\text{train}} = 80$ models for estimation and classification; rest held out. MDS (8 components) + LDA. Averaged over 100 random splits.

4 Results

Stepwise dominates at small m : at $m=2$, stepwise error is 10% (motivating) and 11% (system prompt) vs. 28% for

uniform. Estimated signal is intermediate, confirming the GMM partition captures real structure. On the system prompt task, stepwise at $m=5$ matches uniform at $m=50$ — a 10 $\times$ reduction in per-model query cost.